

Technical Walkthrough – Evidence Package

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Meeting Details

Date: 11th August 2026

Audience:

Calvin (Business SME), Divy and Priya (Technical SME's), Dan (Senior Finance Manager and Project Sponsor)

Purpose

The purpose of this session was to provide stakeholders with an end-to-end technical walkthrough of the scenario modelling framework developed during the project. The session demonstrated how the solution integrates enterprise data sources, controlled business inputs, Excel, Power BI and Python to support scenario configuration, processing, validation and reporting.

The walkthrough also provided an opportunity to explain and challenge the technical design decisions underpinning the prototype, including the choice of technology stack, data architecture, modular Python engine, validation approach and user interface. Stakeholder questions and discussion were used to identify limitations of the current prototype and consider how the architecture could evolve if the solution were subsequently industrialised.

Meeting Notes

Part 1: End-to-End Scenario Modelling Workflow [03:56]

I provided a comprehensive walkthrough of the upstream scenario modelling framework, detailing the user workflow from data input refresh, scenario configuration, and operation selection, to running the Python engine and refreshing outputs, with contributions from Divyratn and Calvin regarding technical aspects and user experience.

User Workflow Overview: I explained the step-by-step process for users, starting with refreshing data inputs, configuring scenarios based on imported data, making inclusion decisions for each case, and selecting operations such as delay, accelerate, farm in, farm down, truncate, and percentage changes. Users then return to the homepage to run the engine and refresh outputs, ensuring updated results for further analysis.

Scenario Operations and Configuration: The meeting covered the types of operations available for scenario modelling, including custom Python functions for manipulating case profiles. I described how users can apply operations like delay, accelerate, farm in, farm down, truncate before/after, and adjust production, OPEX, or CAPEX by percentage or absolute values, with helper text guiding expected inputs and optional parameters.

Python Engine Execution: I detailed the execution of the scenario engine, which processes user-configured operations through modular Python code within Excel. The engine validates each step, provides feedback via green ticks, and ensures that only valid cases and operations are processed, with error handling and validation messages to support troubleshooting.

Output Refresh and Dashboard Integration: After running the engine, users refresh outputs to update the Excel data model, which feeds into Power Pivot dashboards for analysis. I highlighted the importance of validation messages and timestamp checks to ensure outputs are current, and described the integration of scenario results into pivot tables and dashboard visuals for user-driven reporting.

Screenshots Shared

Upstream Scenario Modelling Framework

Home Page

Home

S1

S2

S3

S4

S5

Results

Help

Application Actions

Refresh Inputs

Update source data including model cases and financial profiles

Last refreshed: 10 Aug 26 09:10:37

Run Scenario Engine

Execute scenario operations and generate modelling outcomes

Last run: 11 Aug 26 13:33:12

Refresh Outputs

Refresh modelling results into Pivotables and dashboard visuals

Last refreshed: 11 Aug 26 13:33:35

Outputs includes latest scenario engine results

Follow the workflow above: refresh inputs → configure scenarios (S1-S5) → run the scenario engine → refresh outputs before proceeding with analysis.

Scenario Configuration

Scenario (S1)

All data for testing

7 (of 10) Cases Included | 10 Operations Configured | Distinct operations used: Delay [1], Accelerate [1], Farm In (Increase) [1], Farm Down (Dilution)

Scenario not ready

Scenario (S2)

Empty scenario

Select some cases to make a scenario

Scenario (S3)

Empty scenario

Select some cases to make a scenario

Scenario (S4)

Empty scenario

Select some cases to make a scenario

Scenario (S5)

Empty scenario

Select some cases to make a scenario

Model Engine Log

Run duration: 6.55 seconds

Show Technical Sheets

Hide Technical Sheets

Case listing loaded and validated successfully.

Portfolio case definitions are ready for scenario modelling.

10 valid rows retained from 10 imported.
0 invalid records removed.
0 rows contain incomplete hierarchy attributes.
10 unique case IDs and 10 unique case names validated.

Financial profiles loaded and validated successfully.

Financial profiles for 10 cases are ready for scenario modelling across the 2026–2040 planning horizon.

10 valid rows retained from 10 imported.
0 invalid records removed.
0 rows contain incomplete hierarchy attributes.
10 unique case IDs and 10 unique case names validated.

Scenario instructions prepared successfully.

Scenario instructions have been validated and prepared for execution across all configured scenarios.

50 instruction rows processed.
3 operations on excluded cases cleared.
0 blank or invalid operations defaulted to NoChange.
0 stale parameter sets reset.

Scenario engine completed successfully.

The modelling engine processed 7 case operations and generated updated scenario results.

1 scenarios completed; 4 skipped.
7 case operations processed.
2,310 rows generated across 6 cases.
0 cases have incomplete hierarchy attributes.

Upstream Scenario Modelling Framework

Scenario Configuration

Home

S1

S2

S3

S4

S5

Results

Help

Scenario Name

All data for testing

Description

Last Updated

Configured By

Configure a scenario by selecting planning cases to include or exclude and applying operations that modify timing, scale or input values to evaluate alternative outcomes.

Case Information

Case name, Segment, Region, Country reference data and current Plan Type

Locked Cells

Input Cells

Case ID

Case Name

Segment

Region

Country

Plan Type

Amount 1

Amount 2

Amount 3

Included (y/n)

Choose Operation

Operation Description

Input 1 Label

Input 1 Required

1

Project Golden Dune

G&LC

Asia Pacific

AsPac - Australia

Manual

-600

-600

-600

Delay

Yes to delay (whole number)

Yes

2

Project Sandpiper

G&LC

India

India

Manual

-600

-600

-600

Accelerate

Yes to accelerate (whole number)

Yes

3

Project Frontier West

G&LC

MENA - G&LCE

North Africa - Egypt

Manual

-600

-600

-600

Farm In (Increase)

Increase % (as fraction)

Yes

4

Project Crimson Basin

G&LC

Trinidad

Trinidad

Manual

-600

-600

-600

Farm Down (Dilution)

Dilution % (as fraction)

Yes

5

Project Deepstone

P&O

AGT

AGT - Azerbaijan

Manual

-600

-600

-600

Yes

Truncate Before

Removes values before year

Retain profile from year

Yes

6

Project Phoenix Recovery

P&O

BPX Energy

BPX

Manual

-600

-600

-600

Yes

Truncate After

Removes values before year

Retain profile from year

Yes

7

Project Neptune Expansion

P&O

Gulf of America and Canada

US - GoA

Manual

-600

-600

-600

Yes

Production Change %

Adjusts production by a %

% change (as fraction)

Yes

8

Project Falcon Ridge

P&O

Latin America

Argentina

Manual

-600

-600

-600

Yes

OPEX Change %

Adjusts OPEX by a %

% change (as fraction)

Yes

9

Project Northern Horizon

P&O

MENA - OIL

Middle East - Iraq

Manual

-600

-600

-600

Yes

CAPEX Change %

Adjusts CAPEX by a %

% change (as fraction)

Yes

10

Project Blackwater

P&O

North Sea

North Sea - UK

Manual

-600

-600

-600

Yes

CAPEX Change \$m

Adjusts CAPEX by \$m

Change Y1 (x \$m)

Yes

11

No case

-

-

-

-

-

-

-

12

No case

-

-

-

-

-

-

-

13

No case

-

-

-

-

-

-

-

14

No case

-

-

-

-

-

-

-

1 Case Inclusion

7 / 10

Some cases not configured

2 Operations Inputs

2 / 10

Review configuration

3 Scenario Readiness

45%

Scenario not ready

Data Overview

Configure results using the drop-down selections. Values shown

Average \$Y

Average \$Y

Average \$Y

Cash Capex

Cash Capex

Cash Capex

Operation

Choose whether to include or exclude a case, then choose an operation to apply

Included (y/n)

Choose Operation

Operation Description

Input 1 Label

Input 1 Required

Part 2: Technical Architecture and Data Integration [08:56]

I outlined the technical architecture underpinning the scenario modelling tool, including data sourcing from SAP WRO financial cubes, Azure Data Lake and Data Hub, semantic modelling in Power BI, and integration with Excel via DAX-based OLE DB queries, with input from Divyratn on Python integration and Calvin on user requirements.

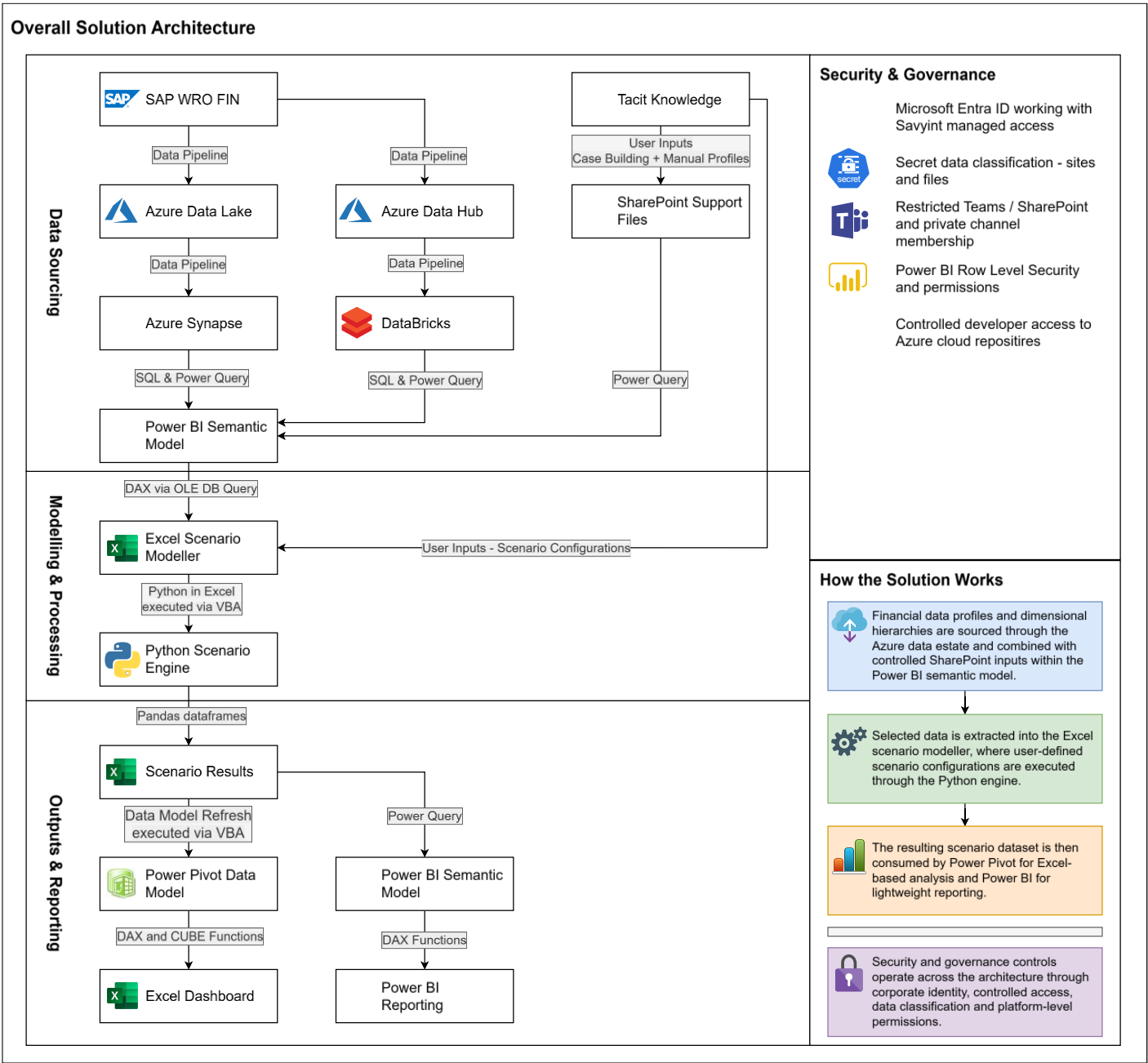
Data Sourcing and Pipeline: I described how transactional and dimensional data is sourced from SAP WRO financial cubes, Azure Data Lake, and Azure Data Hub, with SQL and Power Query used for extraction and minimal transformation. The strategic pathway is to migrate towards Azure Data Hub, with Synapse providing access and Power Query handling joins and filtering.

Semantic Model and Controlled Inputs: The semantic model in Power BI serves as the container for integrated data, combining system-sourced profiles with controlled SharePoint inputs for account aggregation, tax assumptions, manual profiles, and financial entity-to-case mappings. Tacit knowledge and manual overrides are captured via input schedules, enabling bespoke modelling groups and traceability.

Excel Integration via DAX and OLE DB: I explained the use of DAX-based OLE DB queries to expose semantic model outputs to Excel, forming defined table ranges for case listings, profiles, and tax rates. These tables are converted to Pandas data frames in the Excel Python runtime, allowing scenario processing and validation within the Excel modeller.

Security and Access Control: Security measures include restricted Teams and SharePoint access, Power BI row-level security, and controlled developer access to Azure repositories. Users must authenticate via Microsoft accounts to refresh data, ensuring only authorised personnel can trigger scenario updates and access sensitive information.

Diagrams Shared



Part 3: Python Integration and Modular Engine Design [20:28]

I provided an in-depth explanation of the Python integration within Excel, covering modular engine design, execution flow, validation, error handling, and extensibility, with Divyratn contributing insights on technical discoveries and Calvin's requirements for scenario flexibility.

Modular Python Execution in Excel: I described how Python code is executed in Excel in a modular, cell-by-cell manner, similar to a Jupyter notebook. The VBA code triggers Python execution, switches calculation modes, and ensures staged, chunked processing for validation and performance optimisation.

Validation and Error Handling: Each Python module includes validation steps, returning business, technical, and overall status messages. Errors are captured and reported with specific details, enabling users to identify and correct input mistakes, such as mismatched years or missing metrics, before rerunning the engine.

Operations Library and Extensibility: The operations library consists of extensible Python functions for scenario manipulation, including delay, accelerate, farm in, farm down, truncate, and custom metrics adjustments. Helper functions and documentation strings support optional parameters and performance, with the ability to add new functions as user requirements evolve.

Scenario Loop and Post-Processing: The engine loops through each scenario and case ID, applying user-specified operations and handling post-processing steps such as tax calculations. Validation messages and duration tracking provide feedback on engine performance and completion.

Screenshots Shared

The screenshot displays the Microsoft Excel interface with a scenario analysis table and a Python Editor window. The table, titled "SCENARIO XL ARRAY DEFINITION", lists various scenarios (S1 to S5) and their corresponding parameters (Param1 to Param6). The Python Editor window shows a script for defining the scenario XL array, including a function to prepare the scenario table and a function to apply the scenario to the data.

Scenario	Case Name	Include	Operation	Param1	Param2	Param3	Param4	Param5	Param6
S1	Scenario 1: No Change	TRUE	NoChange	0	0	0	0	0	0
S2	Scenario 2: Accelerate	TRUE	Accelerate	0	0	0	0	0	0
S3	Scenario 3: Delay	TRUE	Delay	0	0	0	0	0	0
S4	Scenario 4: Farm In (Increase)	TRUE	Farm In (Increase)	0	0	0	0	0	0
S5	Scenario 5: Farm Out (Dilution)	TRUE	Farm Out (Dilution)	0	0	0	0	0	0

The screenshot displays the Microsoft Excel interface with a scenario analysis table and a Python Editor window. The table, titled "SCENARIO XL ARRAY DEFINITION", lists various scenarios (S1 to S5) and their corresponding parameters (Param1 to Param6). The Python Editor window shows a script for defining the scenario XL array, including a function to prepare the scenario table and a function to apply the scenario to the data.

Scenario	Case Name	Include	Operation	Param1	Param2	Param3	Param4	Param5	Param6
S1	Scenario 1: No Change	TRUE	NoChange	0	0	0	0	0	0
S2	Scenario 2: Accelerate	TRUE	Accelerate	0	0	0	0	0	0
S3	Scenario 3: Delay	TRUE	Delay	0	0	0	0	0	0
S4	Scenario 4: Farm In (Increase)	TRUE	Farm In (Increase)	0	0	0	0	0	0
S5	Scenario 5: Farm Out (Dilution)	TRUE	Farm Out (Dilution)	0	0	0	0	0	0

Part 5: User Decision Support and Helper Tables [40:25]

I discussed the use of helper tables, XLOOKUP functions, and dynamic selections within Excel to support user decision-making during scenario configuration, with Divyratn recalling the evolution of these features and their impact on transparency and usability.

Helper Tables and Dynamic Lookups: Helper tables provide users with reference data for each case, enabling informed decisions on inclusion and operation selection. Dynamic XLOOKUP functions allow users to filter and view relevant metrics, such as cash CapEx in specific years, supporting scenario screening and configuration.

Transparency and User Guidance: The addition of helper text and validation messages improves transparency, guiding users on required and optional inputs, and clarifying the impact of operations. This evolution from the original model enhances usability for both experienced and new users.

Screenshots Shared

Upstream Scenario Modelling Framework | Scenario 3 Configuration

Home | S1 | S2 | **S3** | S4 | S5 | Results | Help

Scenario Name: [Empty]
Description: [Empty]
Last Updated: [Empty]
Configured By: [Empty]

Configure a scenario by selecting planning cases to **include** or **exclude** and applying operations that modify **timing, scale** or **input values** to evaluate alternative outcomes.

Case Information
Case name, Segment, Region, Country reference data and current Plan Type

Locked Cells | **Input Cells**

Case ID	Case Name	Segment	Region	Country	Plan Type	Amount 1	Amount 2	Amount 3	Included (y/n)	Choose Operation	Operation Description	Input 1 Label
1	Project Golden Dune	G&L	Asia Pacific	AsPac - Australia	Manual	-500	38	434				
2	Project Sandpiper	G&L	India	India	Manual	-500	38	434				
3	Project Frontier West	G&L	MENA - GCC	North Africa - Egypt	Manual	-500	38	434				
4	Project Crimson Basin	G&L	Trinidad	Trinidad	Manual	-500	38	434				
5	Project Deepstone	P&O	AGT	AGT - Azerbaijan	Manual	-500	38	434				
6	Project Phoenix Recovery	P&O	BPX Energy	BPX	Manual	-500	38	434				
7	Project Neptune Expansion	P&O	Out of America and Canada	US - GoA	Manual	-500	38	434				
8	Project Falcon Ridge	P&O	Latin America	Argentina	Manual	-500	38	434				
9	Project Northern Horizon	P&O	MENA - OIL	Middle East - Iraq	Manual	-500	38	434				
10	Project Blackwater	P&O	North Sea	North Sea - UK	Manual	-500	38	434				
11	No case	-	-	-	-	-	-	-				
12	No case	-	-	-	-	-	-	-				

Data Overview - Configure results using the drop-down selections. Values shown represent:

	2026	Average 5Y
Cash Capex	-5,000	380
Total Production (mboed)	4,340	Average 5Y
Pre Tax Cash Flow		

Operation
Choose whether to include or exclude a case, then choose an operation to apply

Operation input 1

Ready | Calculate | Accessibility Investigate

11°C Sunny

09:42 11/06/2024

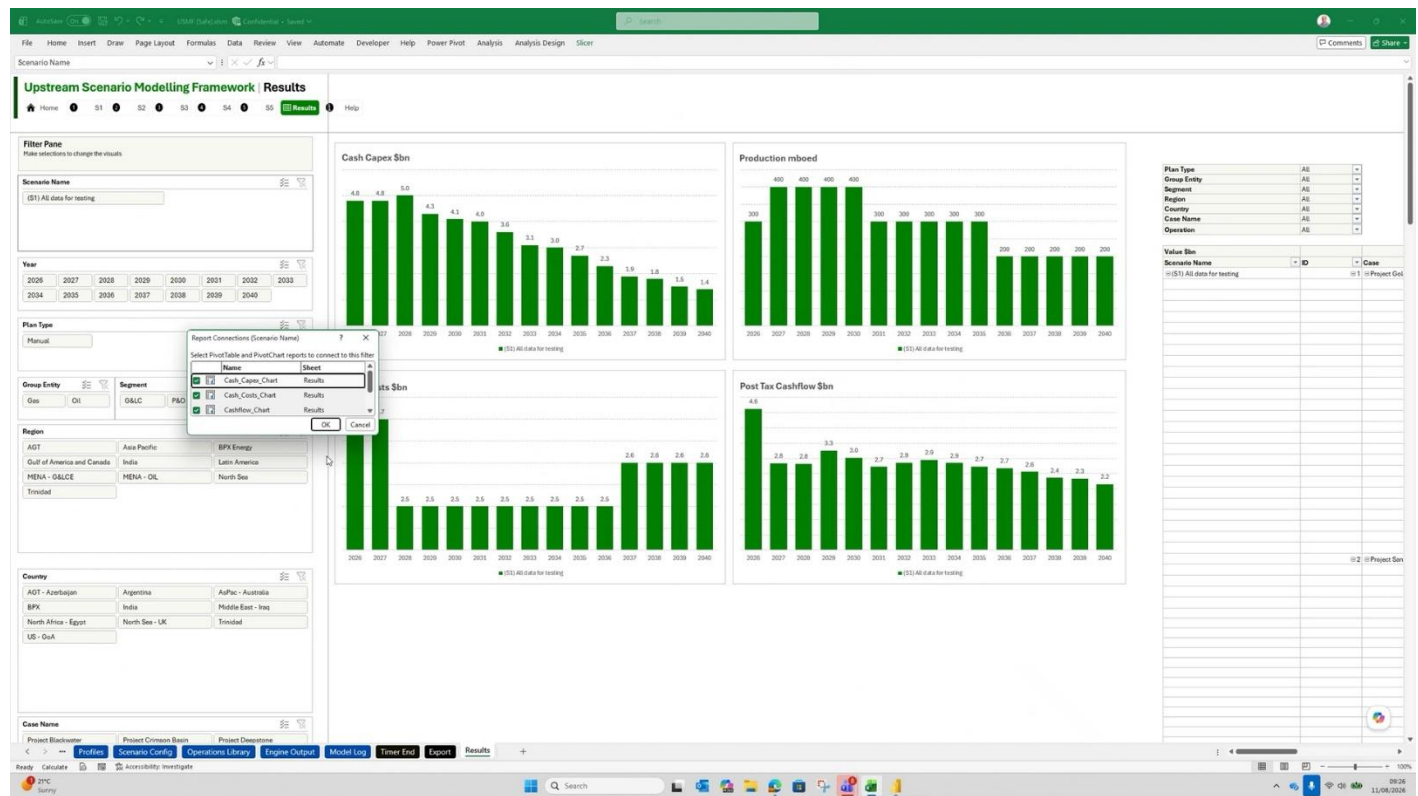
Part 6: Scenario Results, Reporting, and Historical Archiving [34:29]

I explained the process for generating scenario results, integrating them into Excel dashboards and Power BI reports, and archiving historical runs for future reference, with input from Divyratn on reporting flexibility and Calvin's requirements for executive-level responsiveness.

Dashboard and Pivot Table Integration: Scenario results are refreshed and integrated into Power Pivot dashboards within Excel, providing lightweight, interactive reporting capabilities. Users can connect slicers and filters to pivot tables and charts, enabling responsive analysis for executive requests.

Historical Scenario Archiving: I described the semantic model's role as a repository for historical scenario runs, allowing users to archive results from previous submissions and maintain a record of live events in the Excel file. This supports flexible reporting and traceability over time.

Screenshots Shared



Stakeholder Challenge and Architectural Discussion

Following the walkthrough, a stakeholder challenged two aspects of the technical architecture: why the prototype used a Power BI semantic model rather than performing more of the processing within Databricks, and whether a more robust application architecture could remove the dependency on Excel.

I welcomed this challenge as both questions highlighted an important distinction between the architecture selected for a time-constrained prototype and the architecture that might be appropriate if the framework were subsequently industrialised.

Databricks could potentially support a greater proportion of the upstream data preparation, transformation and processing. However, the Power BI semantic model was retained within the prototype because it provided the analytical relationships and business logic required to integrate the underlying datasets, while also aligning with my existing technical capability and the development access available during the project. The transformation logic was intentionally kept relatively lightweight and tabular where possible, providing a potential pathway for more processing to be migrated upstream in future.

Excel was similarly a deliberate prototype choice rather than an assumption about the permanent application architecture. It provided a familiar and flexible environment for Planning users to configure scenarios and perform bespoke analysis. A Python/Streamlit interface was also developed and tested during the project as an alternative, but further development was paused due to project time constraints and stakeholder preference for the Excel-based approach.

For an industrialised implementation, I would therefore recommend reassessing these architectural boundaries. This could include moving additional processing upstream into Databricks and introducing a dedicated application front end, while retaining an export-to-Excel capability to support users requiring flexible, customisable tables, analysis and plotting.

Teams Message Communication

09:07

Apologies for being late

Questions for when you are ready:

Why use the PBI semantic model versus an alternative in DBX?

What would the alternatives be to having Excel in this application at all? Are there more robust alternatives to achieve what this product brings?

10:45

11/08/2026 09:12

Questions for when you are ready: Why use the PBI semantic model versus an alternative in DBX? What would the alternatives be to...

Hi [redacted] thanks for your message. I agree with both points, and these are definitely areas I'd reassess if the prototype were taken forward into a more industrialised solution.

DBX could certainly be used for more of the data transformation and processing. I retained the Power BI semantic model because it provides the analytical relationships and business logic across the datasets, but also pragmatically because of my existing Power BI experience and the fact that I don't currently have the engineering capability/access to develop directly in Databricks.

Excel was intentionally used as the prototype shell due to its familiarity and flexibility for the Planning team. I also tested a Python/Streamlit alternative, but paused this due to EOS and the team's preference to continue with Excel.

For a more industrialised solution, I'd reassess the architecture, potentially moving more processing upstream into DBX and considering a dedicated application front end. However, I would retain an export-to-Excel capability to support users' need for flexible, customisable tables, analysis and plotting.